Optimizing Library Operations Through SQL-Based Automation and Analytics

Presented by: VICTORIA ADEMIJUBE

Problem Statement

- Increase in overdue books while operational processes remains manual which is error-prone and time consuming.
- Delay in updating book availability after returns affects service speed and reduce member satisfaction.
- Lack of real-time visibility into branch performance and staff productivity.
- Limited information to measure success of recent promotional campaigns.

Objectives

Primary objective

- Implement an automated SQL-driven system to improve operational efficiency, service delivery and management decision-making.

Secondary objectives

- Automate identification of overdue books and calculation of fines.
- Enable real-time updates of book availability upon return.
- Generate standardized branch performance reports.
- Identity active members based on borrowing behaviours.
- Measure staff productivity.
- Detect patterns in damaged book returns.

Dataset Overview

- The dataset include 6 tables:
- Books- Book information , 1000 rows and 7 columns
- Members- Registered library users , 4000 rows and 4 columns
- Isssued_status- Book issuance records , 420 rows and 6 columns
- Return_status- Book return records, 150 rows and 6 columns
- Employees- Staff and branch assignment , 40 rows and 5 columns
- Branch- library branch information , 5 rows and 4 columns

Methodology

Tools Used:

- SQL

Data cleaning

- Data set are clean

Data Insights

	member_id [PK] character varying (7)	overdue_days integer	fine numeric
1	C1297	706	353.0
2	C2116	702	351.0
3	C1400	699	349.5
4	C2757	698	349.0
5	C1806	696	348.0
6	C1582	688	344.0
7	C1646	678	339.0
8	C1084	677	338.5
9	C3247	673	336.5
10	C2992	668	334.0

- Member with id C1297 has highest number of days overdue (706 days) and is to pay a fine of $353.0

Data Insights

	issued_book_name character varying (100)	issued_date date	return_date date	book_quality character varying (10)	availability text
1	Monitored Human-Resource Definition	2024-07-08	[null]	[null]	unavailable
2	Implemented Fresh-Thinking Architecture	2024-07-31	2024-09-18	Fair	available
3	Multi-Channeled Dedicated Intranet	2024-02-10	2024-03-24	Good	available
4	Synergistic Needs-Based Pricing Structure	2025-04-21	2025-06-05	Good	available
5	Cross-Group 4Thgeneration Secured Line	2024-07-31	2024-09-16	Good	available
6	Organized Holistic Knowledgebase	2025-04-21	[null]	[null]	unavailable
7	Up-Sized Asynchronous Product	2024-04-05	[null]	[null]	unavailable

- This shows whether an issued book has been returned and the condition of returned books.
- “Monitored Human Resource Definition” is currently unavailable while “Implemented Fresh Thinking Architecture is available”.

Data Insights

	branch_id [PK] character varying (7)	total_books_issued bigint	total_book_returned bigint	revenue_generated numeric
1	B001	105	37	744.67
2	B002	101	36	655.05
3	B005	71	27	538.09
4	B003	78	30	523.11
5	B004	65	20	424.45

- Branch 001 issued the highest number of books and generated the largest revenue.

Data Insights

	member_id character va	member_name character varying (	issued_date date	member_ text
1	C1871	Elijah Rodriguez	2025-12-08	active
2	C789	Melanie Walters	2025-12-06	active
3	C3718	Kevin Ponce	2025-11-12	active
4	C2113	David Williams	2025-11-14	active
5	C654	David Robles	2025-11-09	active
6	C1265	John Washingt...	2025-11-22	active
7	C2929	William Brown	2025-12-04	active
8	C307	Riley Martinez	2025-11-16	active
9	C212	Catherine Burch	2025-11-10	active
10	C213	Danielle Reid	2025-11-10	active
11	C3750	Marcus Lopez	2025-12-08	active
12	C3670	Ronald Kennedy	2025-12-04	active
13	C122	Elizabeth Clark	2025-12-04	active
14	C114	Taylor Mathis Jr.	[null]	inacti...
15	C115	Stephanie Gard...	[null]	inacti...
16	C116	Justin Baxter	[null]	inacti...

- Presently, here are 13 members who have borrowed at least a book in the last 60 days.

Data Insights

	emp_id [PK] character var	branch_id character v	total_books bigint
1	E127	B005	18
2	E117	B005	17
3	E132	B004	17
4	E110	B003	17
5	E134	B005	16
6	E129	B003	15
7	E130	B001	15
8	E108	B004	14
9	E133	B003	14
10	E126	B004	14
11	E111	B002	13

- Employees E127 and E117 issued the highest number of books, 18 and 17 respectively.
- They both work in branch B005

Data Insights

	member_id [PK] character varying (7)	count bigint

	member_id [PK] character varying (7)	count bigint
1	C2605	1
2	C1326	1
3	C1042	1
4	C4072	1

Total rows: 22 Query complete 00:00:00.247

- There was no member who returned damaged books more than once.
- However, 22 members returned books which were damaged once.

Data Insights

	book_title character varying (100)	issued_date date	return_date date	availability text
1	Monitored Human-Resource Definition	2024-07-08	[null]	unavailable
2	Implemented Fresh-Thinking Architecture	2024-07-31	2024-09-18	available
3	Multi-Channeled Dedicated Intranet	2024-02-10	2024-03-24	available
4	Synergistic Needs-Based Pricing Structure	2025-04-21	2025-06-05	available
5	Cross-Group 4Thgeneration Secured Line	2024-07-31	2024-09-16	available
6	Organized Holistic Knowledgebase	2025-04-21	[null]	unavailable
7	Up-Sized Asynchronous Product	2024-04-05	[null]	unavailable
8	Re-Engineered Background Project	2025-08-16	[null]	unavailable
9	Phased Next Generation Success	2024-08-12	2024-09-29	available
10	Re-Contextualized Clear-Thinking Definition	2024-11-13	[null]	unavailable
11	Diverse Coherent Hardware	2025-08-04	2025-09-19	available
12	Decentralized Transitional Knowledgebase	2024-03-10	2024-04-12	available

Recommendations

- Implement automated overdue alerts and fine escalation rules.
- Replicate best practices from high-performing branches.
- Educate members who return damaged books on proper book handling.
- Launch targeted re-engagement campaigns for inactive members.
- Recognize top performing staff through awards or bonuses.

THANK YOU